VoxHub CM4 — Home Assistant voice hub
USB-C PD (9–20 V) -> 5 V buck -> CM4 Lite + USB hub + audio + Zigbee

Power: USB-C PD sink, 5 V buck

File: power.kicad_sch

Compute Module 4 socket, microSD, user interface

File: cm4.kicad_sch

USB 2.0 hub and USB-A host ports

File: usb.kicad_sch

Audio: PDM microphones, PDM-to-I2S, class-D speaker amplifier

File: audio.kicad_sch

Zigbee / Thread radio (RF-BM-2652P2, CC2652P)

File: zigbee.kicad_sch

H1
M2.5 board mount

H2
M2.5 board mount

H3
M2.5 board mount

H4
M2.5 board mount

FID1
Fiducial

FID2
Fiducial

FID3
Fiducial

H5
CM4 spacer M2.5x1.5

GND

H6
CM4 spacer M2.5x1.5

GND

H7
CM4 spacer M2.5x1.5

GND

H8
CM4 spacer M2.5x1.5

GND

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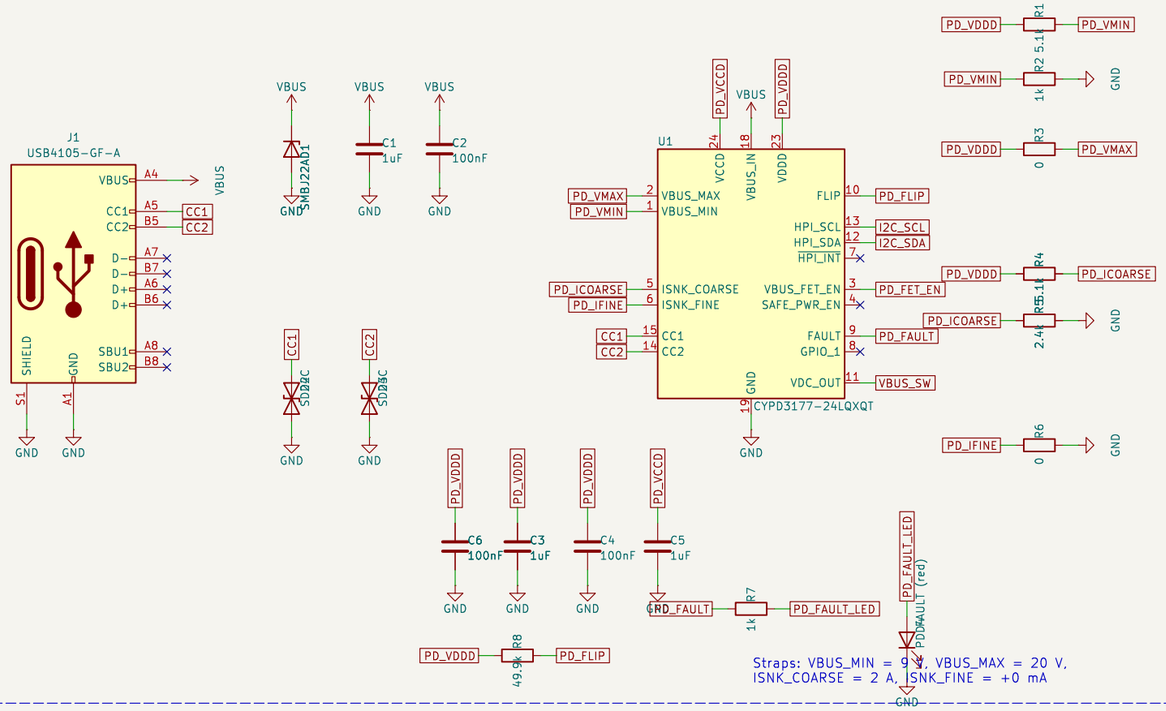
VoxHub (open hardware)

Sheet: /
File: voxhub.kicad_sch

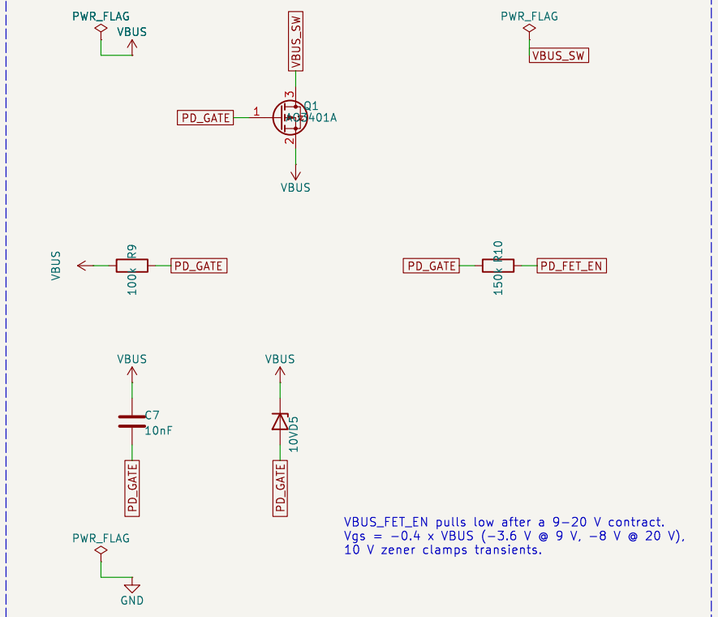
Title: VoxHub CM4 — Home Assistant voice hub carrier

Size: A3	Date: 2026-11-12	**Rev: A2**
KiCad E.D.A. 9.0.9		Id: 1/6

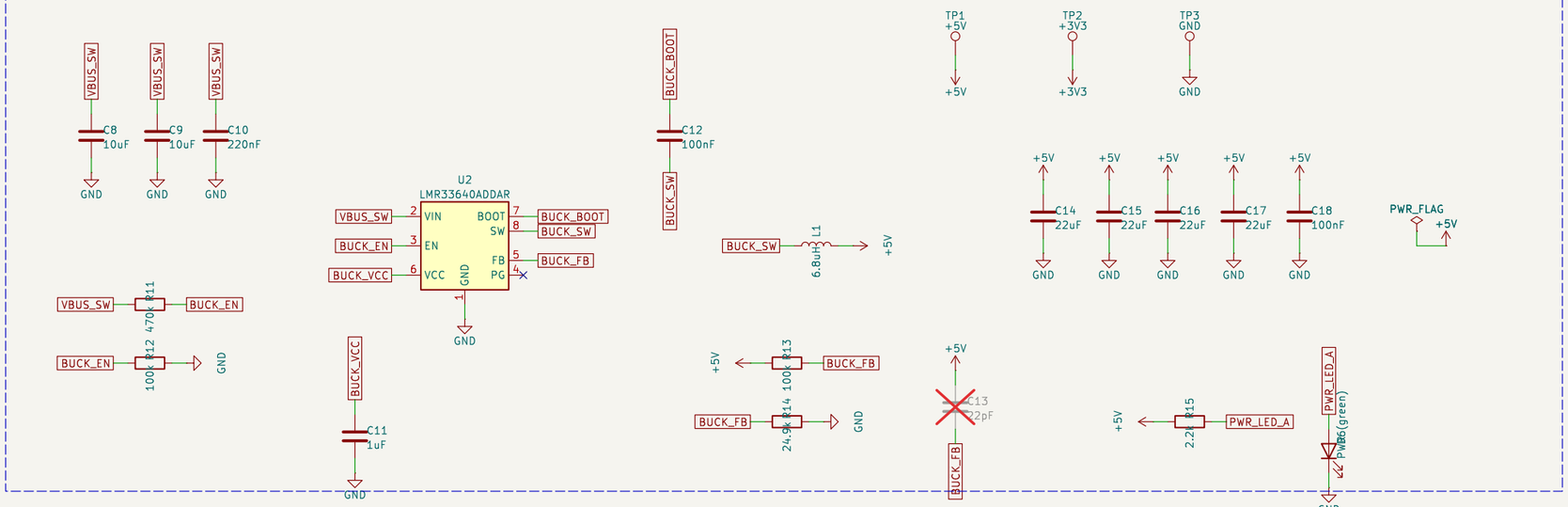
USB-C input and PD sink controller (CYPD3177 'BCR', resistor-strapped: 9–20 V, 2 A minimum)



VBUS load switch: P-FET driven by the BCR's active-low, slew-controlled Vgate driver



4 V synchronous buck (LMR33640, 400 kHz) — UVLO 7.0 V keeps it off at vSafe5V



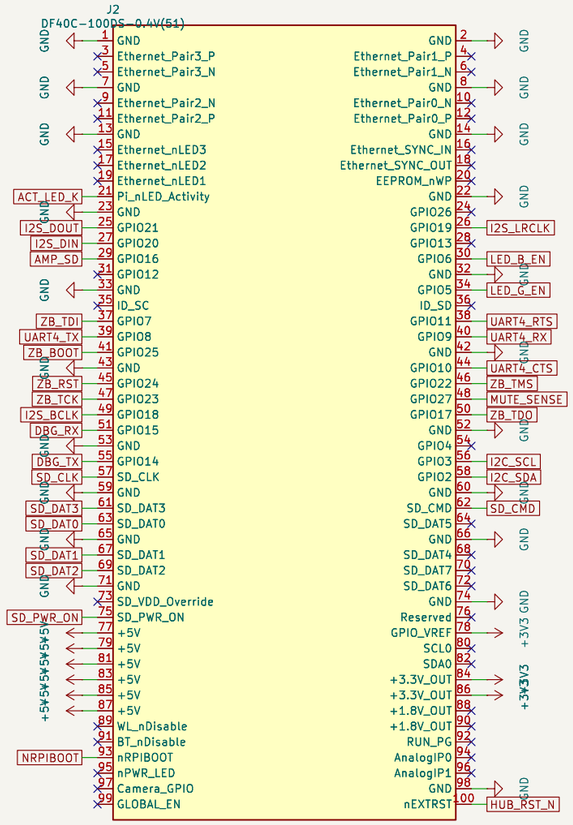
Generated by hardware/gen — do not edit by hand

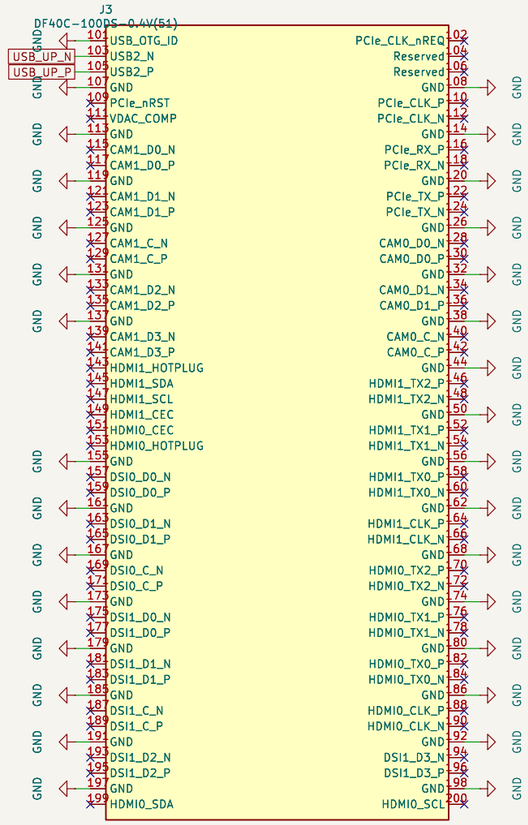
VoxHub (open hardware)

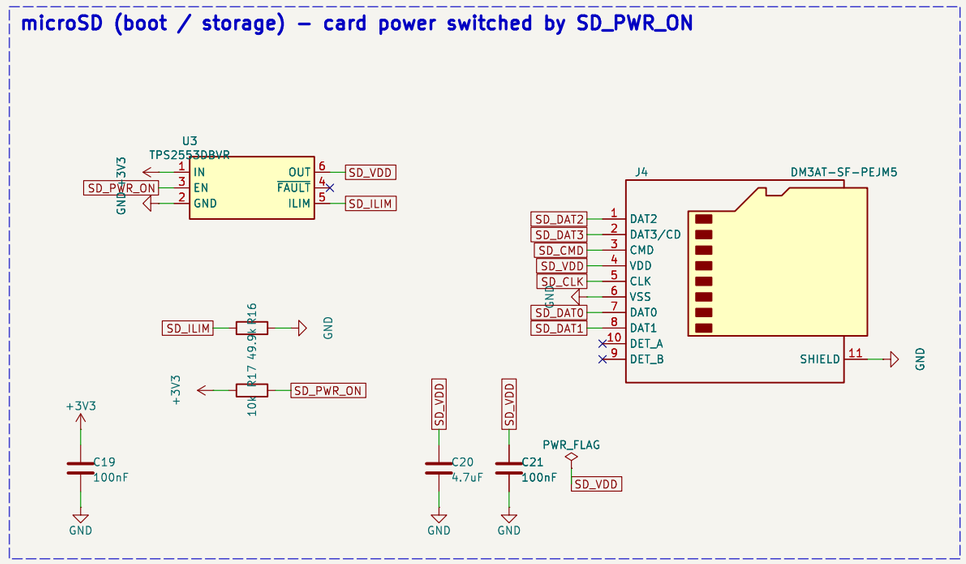
Sheet: /Power: USB-C PD sink, 5 V buck/
File: power.kicad_sch

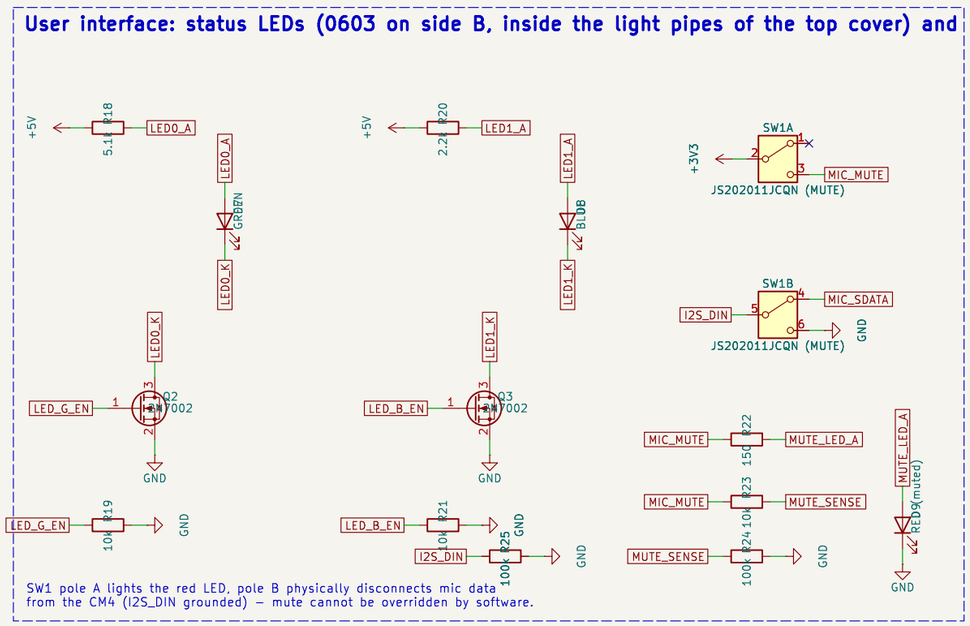
Title: Power: USB-C PD sink, 5 V buck

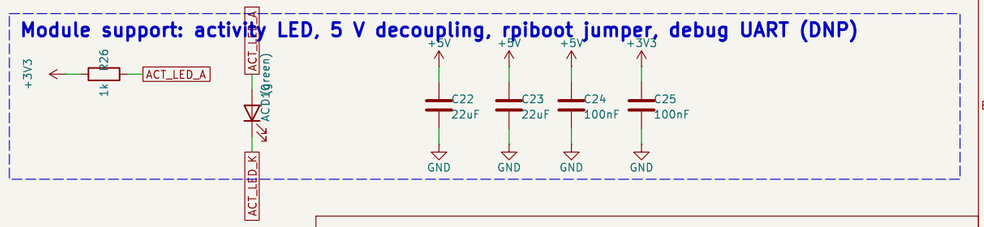
Size: A3	Date: 2026-11-12	**Rev: A2**
KiCad E.D.A. 9.0.9		Id: 2/6

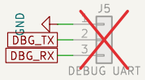














Module: Raspberry Pi CM4 Lite, wireless (CM4104000 / CM4102000 / CM4108000)
GPIO map: see README. USB_OTG_ID tied low = host.

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VoxHub (open hardware)

Sheet: /Compute Module 4 socket, microSD, user interface/
File: cm4.kicad_sch

Title: Compute Module 4 socket, microSD, user interface

Size: A3	Date: 2026-11-12	**Rev: A2**
KiCad E.D.A. 9.0.9		Id: 3/6

USB2514B hub: upstream = CM4 USB2, ports 1–2 = USB-A, ports 3–4 disabled by strap

U4
USB2514B-I/M2

RESET_N is driven by the CM4 (nEXTRST, pin 100); low during reset, high once the SoC boots.

VBUS_DET
USBDM_UP
USBDP_UP
RESET_N
TEST
RBIAS
XTALIN/CLKIN
XTALOUT
PLLFILT
CRFILT
SCL/SMBCLK/CFG_SEL0
SDA/SMBDATA/NON_REM1
SUSP_IND/LOCAL_PWR/NON_REM0
HS_IND/CFG_SEL1
VDDA33
VDD33
VSS
USBDM_DN1/PRT_DIS_M1
USBDP_DN1/PRT_DIS_P1
PRTPWR1/BC_EN1
OCS_N1
USBDM_DN2/PRT_DIS_M2
USBDP_DN2/PRT_DIS_P2
PRTPWR2/BC_EN2
OCS_N2
USBDM_DN3/PRT_DIS_M3
USBDP_DN3/PRT_DIS_P3
PRTPWR3/BC_EN3
OCS_N3
USBDM_DN4/PRT_DIS_M4
USBDP_DN4/PRT_DIS_P4
PRTPWR4/BC_EN4
OCS_N4

HUB_RBIAS R27 12k GND
HUB_PLLFILT C28 100nF GND
HUB_CRFILT C29 100nF GND
HUB_VBUSDET R28 100k +3V3
HUB_CFG0 R29 100k GND
HUB_CFG1 R30 100k GND
HUB_NONREM0 R31 100k GND
HUB_NONREM1 R32 100k GND
HUB_RST_N GND
Y1 24MHz CL12pF HUB_XI HUB_XO GND
HUB_XI C26 15pF GND
HUB_XO C27 15pF GND
C30 100nF C31 100nF C32 100nF C33 100nF C34 100nF C35 100nF C36 4.7uF +3V3 GND
R33 10k +3V3 HUB_DIS3M
R34 10k +3V3 HUB_DIS3P
R35 10k +3V3 HUB_DIS4M
R36 10k +3V3 HUB_DIS4P

Downstream ports: per-port 1.2 A current-limited switch, ESD protection, stacked USB-A

U5
TPS2553DBVR
IN EN GND OUT FAULT ILIM
USB_PWR1 +5V GND
USB1_VBUS USB_OC1_N USB1_ILIM

U7
USB4105-2SC6
USB1_N USB1_P USB1_VBUS GND

USB1_ILIM R37 20k GND
+3V3 R38 10k USB_OC1_N

C37 4.7uF +5V GND
C38 100uF USB1_VBUS GND
C39 100nF USB1_VBUS GND
PWR_FLAG USB1_VBUS

J6
61400826021
VBUS1 VBUS2 D1+ D1- D2+ D2- Shield GND1 GND2
USB1_VBUS USB2_VBUS USB1_P USB1_N USB2_P USB2_N GND

U6
TPS2553DBVR
IN EN GND OUT FAULT ILIM
USB_PWR2 +5V GND
USB2_VBUS USB_OC2_N USB2_ILIM

U8
USB4105-2SC6
USB2_N USB2_P USB2_VBUS GND

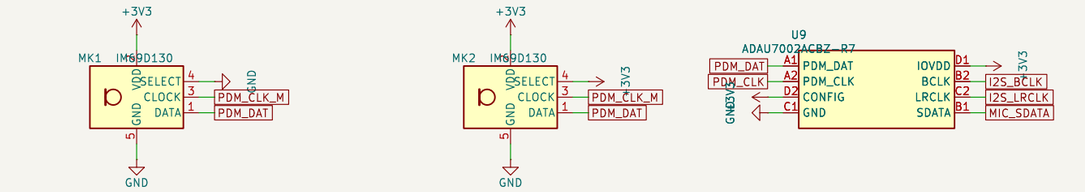
USB2_ILIM R39 20k GND
+3V3 R40 10k USB_OC2_N

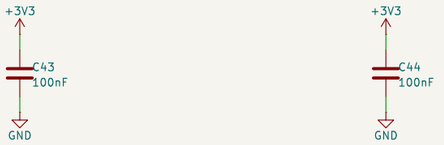
C40 4.7uF +5V GND
C41 100uF USB2_VBUS GND
C42 100nF USB2_VBUS GND
PWR_FLAG USB2_VBUS

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VoxHub (open hardware)
Sheet: /USB 2.0 hub and USB-A host ports/
File: usb.kicad_sch
Title: USB 2.0 hub and USB-A host ports

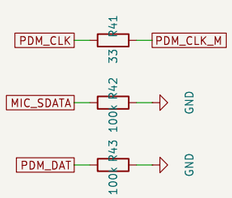
Size: A3	Date: 2026-11-12	**Rev: A2**
KiCad E.D.A. 9.0.9		Id: 4/6

Stereo MEMS microphone array (bottom port through PCB) and PDM -> I2S converter





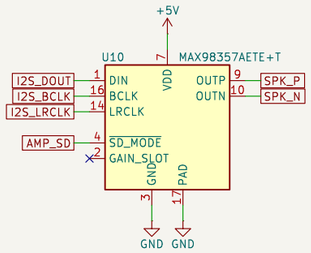
MK1 SELECT=GND (left), MK2 SELECT=VDD (right); 70 mm apart

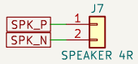


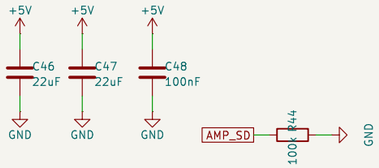


CONFIG = IOVDD -> I2S output; BCLK = 64 fs, PDM clock = 64 fs (3.072 MHz @ 48 kHz)

Speaker amplifier (MAX98357A, I2S in, 3.2 W into 4 ohm from 5 V)







SD_MODE driven by GPIO16 (3.3 V = left channel, low = shutdown). GAIN_SLOT open = 9 dB.

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VoxHub (open hardware)

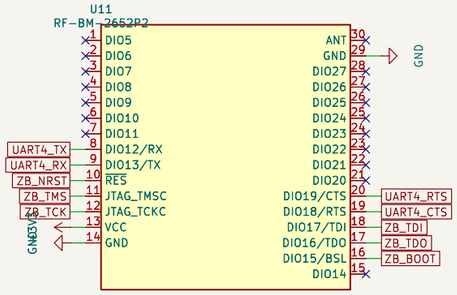
Sheet: /Audio: PDM microphones, PDM-to-I2S, class-D speaker amplifier/

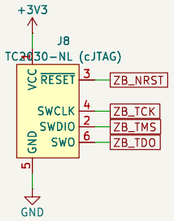
File: audio.kicad_sch

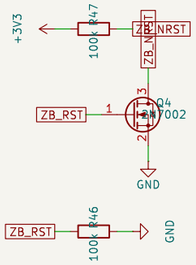
Title: Audio: PDM microphones, PDM-to-I2S, class-D speaker amplifier

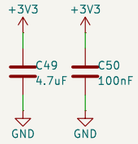
Size: A3	Date: 2026-11-12	**Rev: A2**
KiCad E.D.A. 9.0.9		Id: 5/6

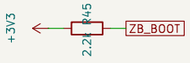
CC2652P network co-processor on CM4 UART4 (GPIO8-11) + ROM serial bootloader strap + JTAG











UART: DIO13 = module TX -> GPIO9 (RXD4), DIO12 = module RX <- GPIO8 (TXD4).
DIO18 = module RTS -> GPIO10 (CTS4), DIO19 = module CTS <- GPIO11 (RTS4) [TI LAUNCHXL-CC1352P-2 assignment].
DIO15 = ROM bootloader backdoor, active low (GPIO25). Reset: GPIO24 HIGH pulls RES low (open drain).
JTAG for recovery: TMS = GPIO22, TCK = GPIO23, TDO = GPIO17, TDI = GPIO7 (4-wire, OpenOCD bit-bang).

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VoxHub (open hardware)

Sheet: /Zigbee / Thread radio (RF-BM-2652P2, CC2652P)/
File: zigbee.kicad_sch

Title: Zigbee / Thread radio (RF-BM-2652P2, CC2652P)

Size: A3	Date: 2026-11-12	**Rev: A2**
KiCad E.D.A. 9.0.9		Id: 6/6